

Multi-Agent System — Maze Navigation (GUI)

PROGRAM:

```
import tkinter as tk
from tkinter import messagebox

maze = [
    [1,1,1,1,1],
    [1,0,0,0,1],
    [1,0,1,0,1],
    [1,0,0,0,1],
    [1,1,1,1,1]
]

cs = 60
p = [1,1]
g = [3,3]

r = tk.Tk()
c = tk.Canvas(r, width=5*cs, height=5*cs)
c.pack()

def draw():
    for i in range(5):
        for j in range(5):
            col = "black" if maze[i][j] else "white"
            c.create_rectangle(j*cs, i*cs, (j+1)*cs, (i+1)*cs, fill=col)
    c.create_rectangle(g[1]*cs, g[0]*cs, (g[1]+1)*cs, (g[0]+1)*cs, fill="green")

def player():
    c.delete("p")
    x,y = p[1]*cs+cs//2, p[0]*cs+cs//2
    c.create_oval(x-15, y-15, x+15, y+15, fill="blue", tags="p")

def move(e):
    d = {"Up":(-1,0), "Down":(1,0), "Left":(0,-1), "Right":(0,1)}
    if e.keysym in d:
        nx,ny = p[0]+d[e.keysym][0], p[1]+d[e.keysym][1]
        if maze[nx][ny]==0:
            p[0],p[1]=nx,ny
            player()
        if p==g:
            messagebox.showinfo("Done", "Goal Reached")
            r.destroy()
```

```
draw()  
player()  
r.bind("<Key>", move)  
r.mainloop()
```

OUTPUT:

